

Object Detection

YOLOv8

Anuraj Sapkota

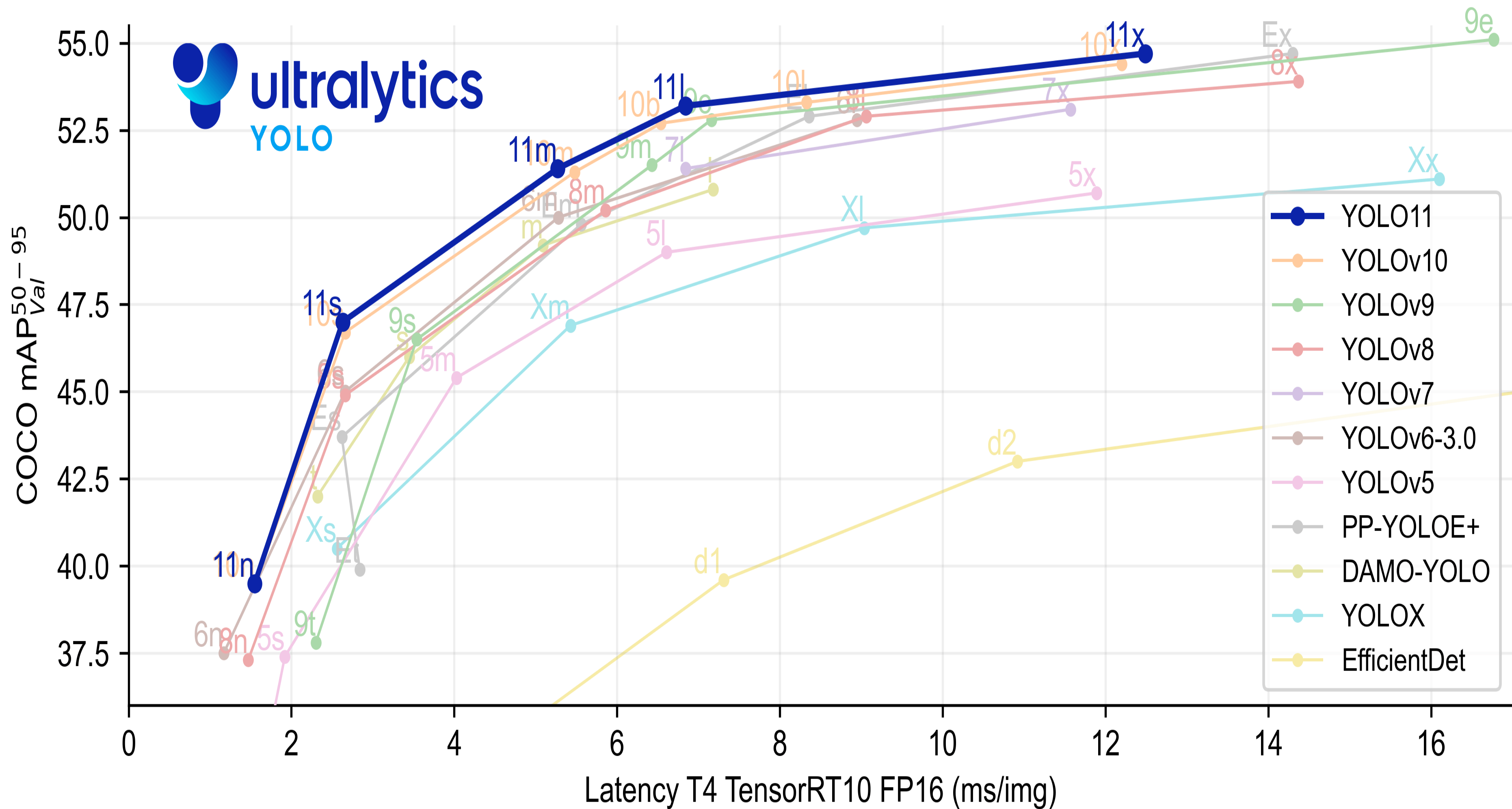
What is Object Detection?

- Object detection is a computer vision task that identifies and locates objects in images or videos.
- Involves drawing bounding boxes around objects and classifying them (e.g., "car," "person").
- Applications: Autonomous vehicles, security surveillance, medical imaging, and robotics.



— YOLO (You Only Look Once)

- Processes the entire image in one NN evaluation.
- Divides image into a grid; predicts bounding boxes and class probabilities per grid cell.
- Extremely fast (real-time on standard hardware), simple architecture.

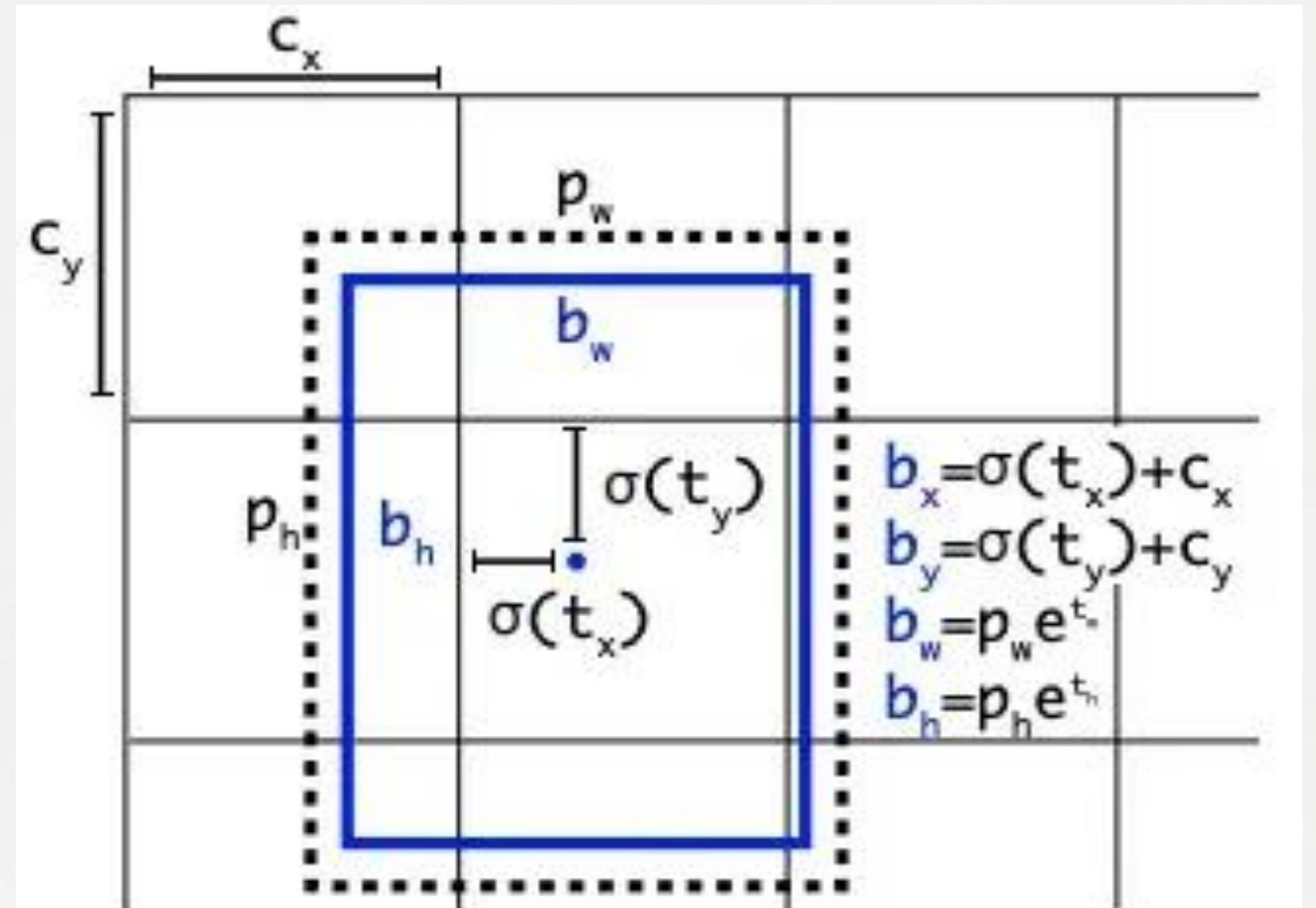


— YOLOv8

- Released by Ultralytics in 2023, latest in the YOLO series (not officially from original authors but highly regarded)
- Supports object detection, instance segmentation, pose estimation, and classification.
- Extremely fast (real-time on standard hardware), simple architecture.

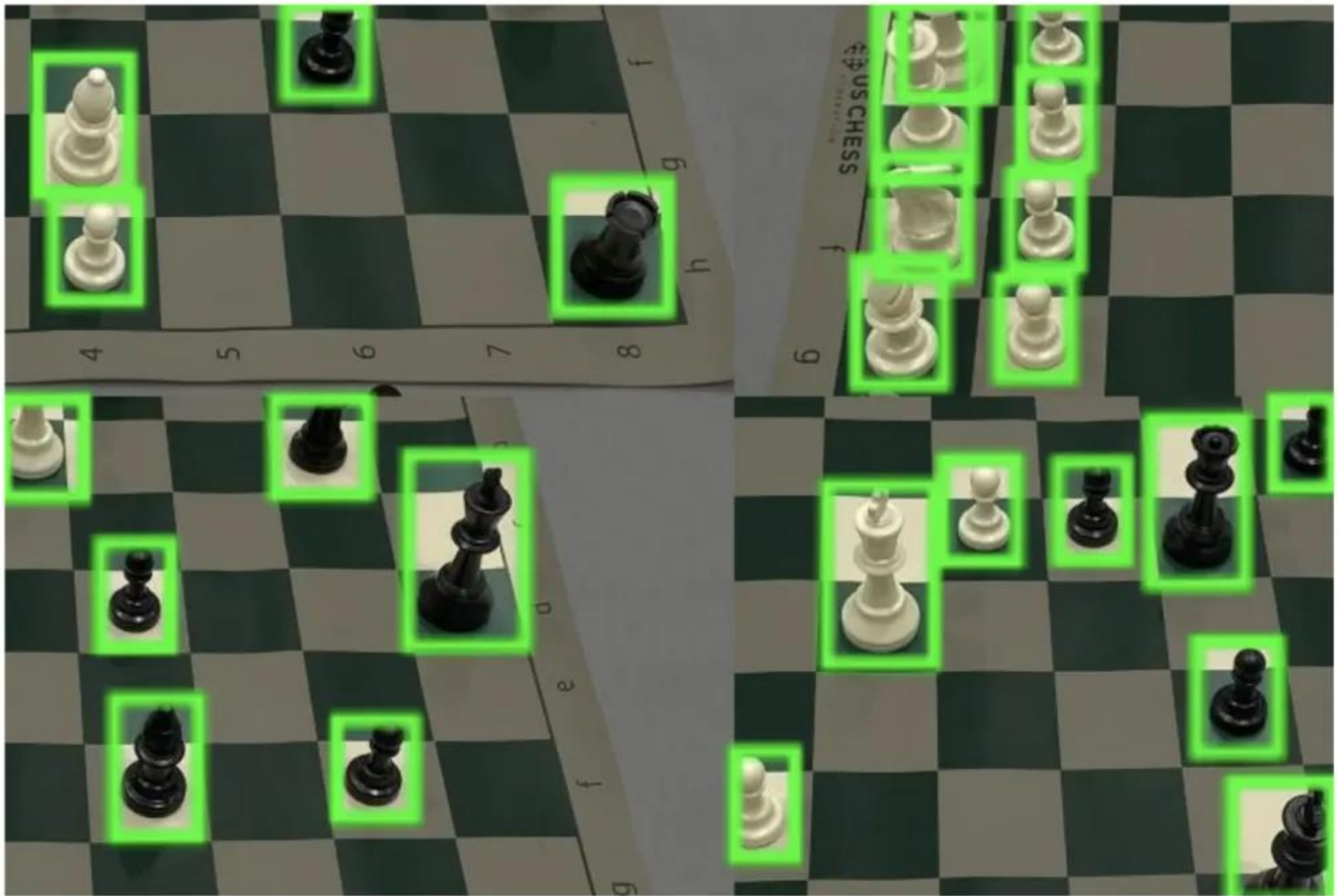
Anchor Free Detection

- YOLOv8 is an anchor-free model. This means it predicts directly the center of an object instead of the offset from a known anchor box.
- Predict the center point of each object directly.
- Predicts the bounding box size from that center
- Predicts the confidence and class for each detected center.



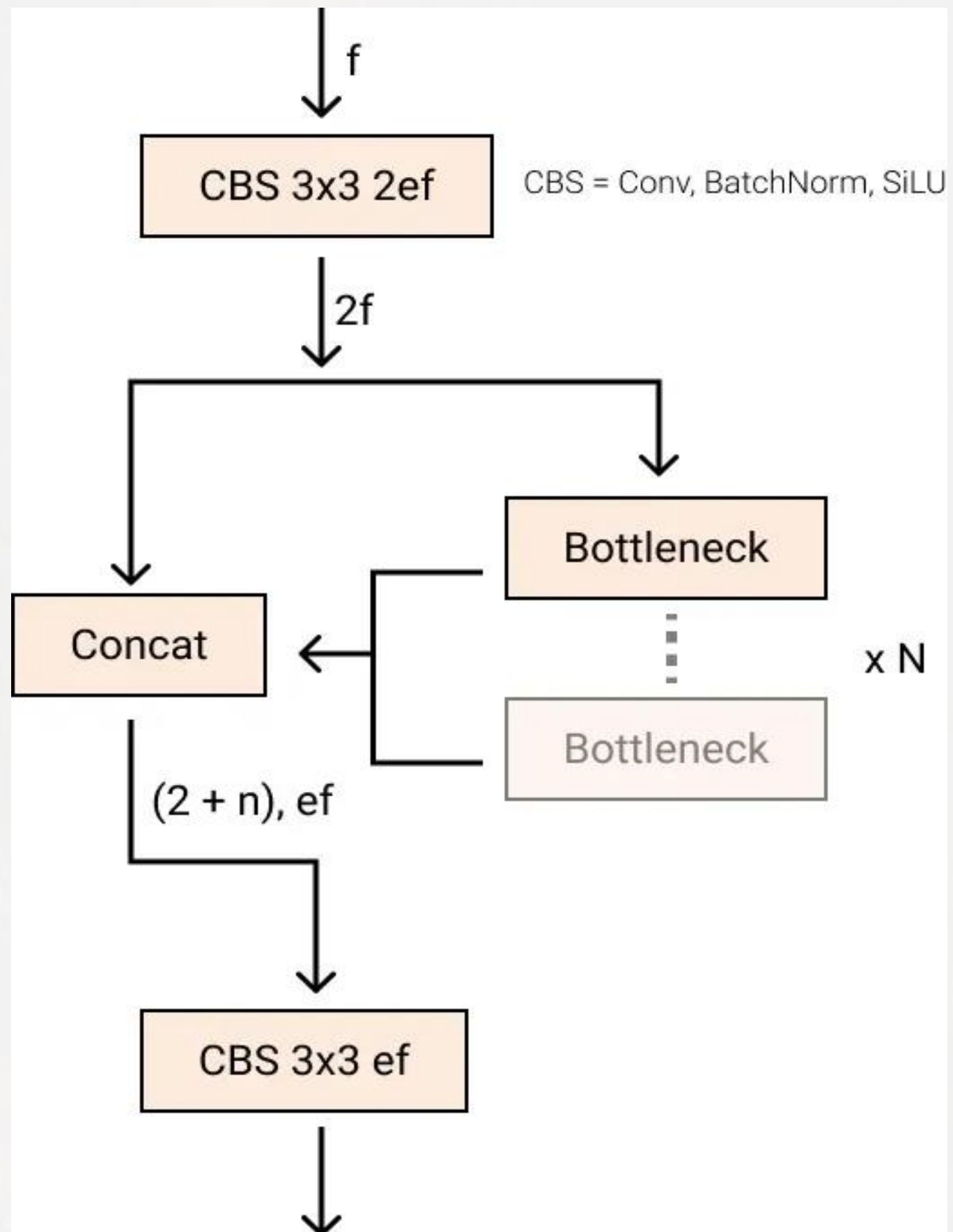
Mosaic Augmentation

- A data augmentation technique in which four different images are combined into single image. Like collage
- Each four images are:
 - randomly scaled,
 - randomly cropped,
 - placed into one of four



New Convolutions

- The stem's first 6x6 conv is replaced by a 3x3, the main building block was changed, and C2f replaced C3 .
- Also has several Bottleneck layers.
- But: instead of keeping just the last output, it concatenates the outputs from all bottlenecks.
- more gradient flow, stronger representation.
- Features are concatenated directly, even if they have different channel dimensions.

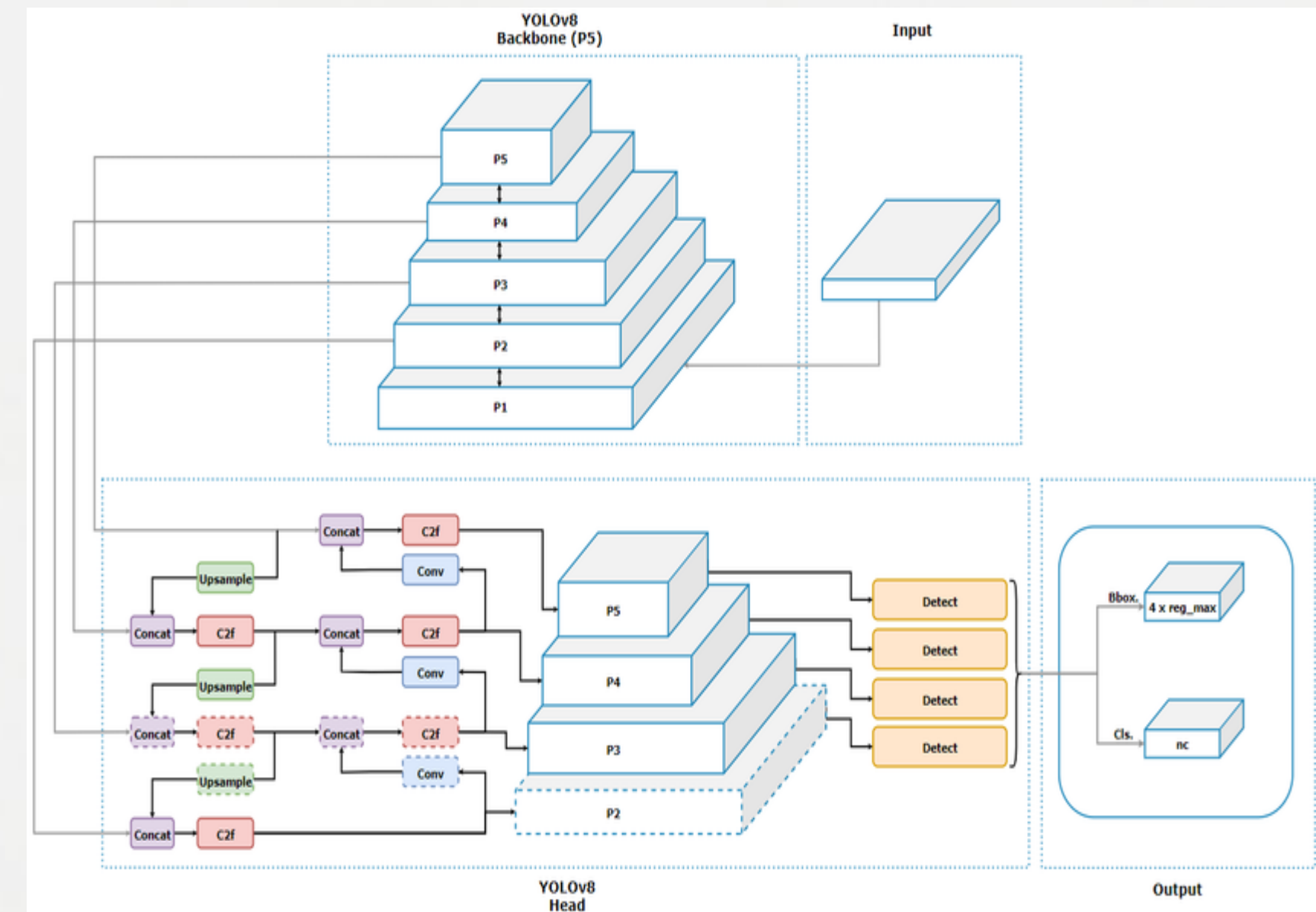


Experiment	Precision/%	Recall/%	$mAP_{0.5}/\%$	Params/M	GFLOPs
YOLOv8n	70.1	68.2	73.3	3.011	8.194
YOLOv8s	70.9	72.9	76.1	11.12	28.4
YOLOv8m	72.4	73.6	76.4	25.85	79.1
YOLOv8l	72.9	73.9	77.8	43.63	165.403
YOLOv8x	73.6	74.3	78.5	68.15	258.123

Architecture

– Backbone

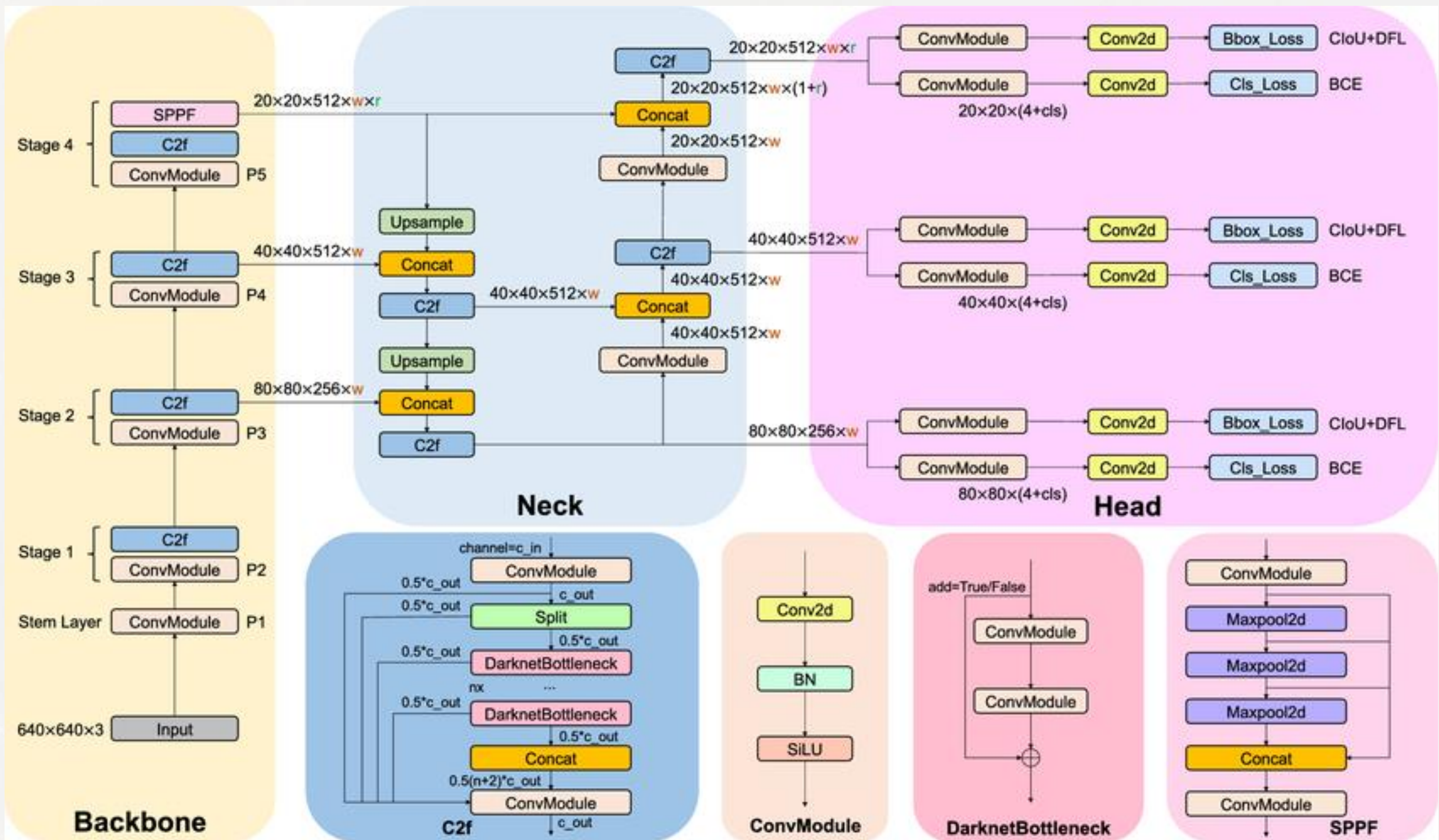
- Uses Conv modules (convolutions + batch norm + SiLU activation).
- C2f module: Replaces C3; combines high-level context with fine-grained features via bottleneck blocks.
- SPFF(Spatial Pyramid Pooling Fast): Aggregates features at different scales for robustness to object sizes.



— Architecture

– Neck

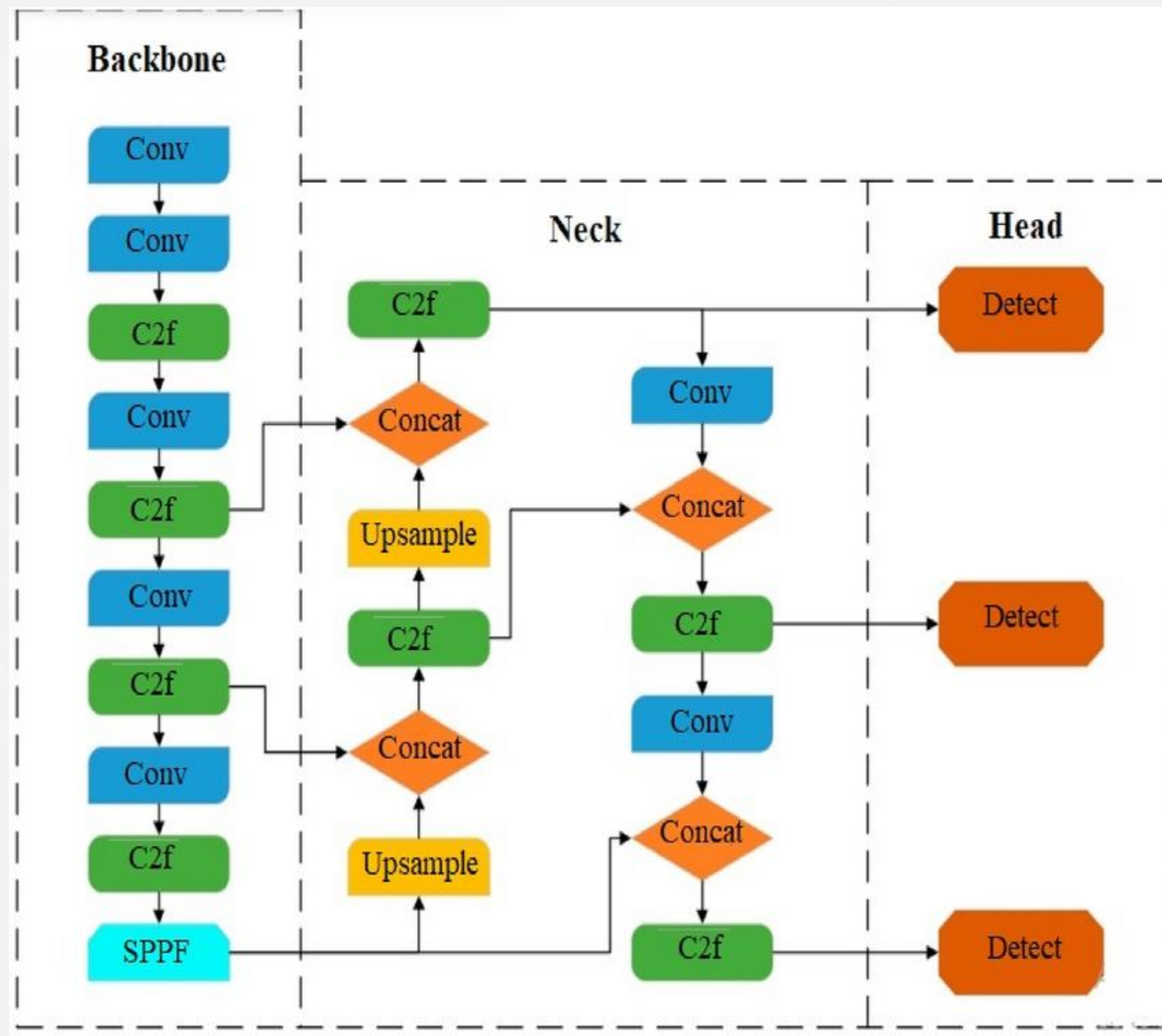
- Connects backbone to head; fuses features from different resolutions.
- Uses Path Aggregation network: bottom-up and top –bottom paths for semantic and spatial info.
- C2f for efficient bottleneck design with split- concat operations.
- Unsampling and concatenation to handle multi-scale objects.
- Improves localization by combining low-level and high-level features.



— Architecture

– Head

- Anchor-free: No predefined anchors; directly predicts center, width, height via regression
- Decoupled design: Separate branches for classification and bounding box regression.
- Outputs: Three detection heads at different scales (P3, P4, P5) for small/medium/large objects.
- Loss functions: Binary Cross-Entropy for classes, Distribution Focal Loss for boxes.



Advantages and Limitations

- **Advantages:**

Real-time speed, high accuracy, easy deployment (PyTorch-based), multi-task support.

- **Limitations:**

May underperform on very small objects; requires labeled data for fine-tuning.